

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1- Concept Mapping

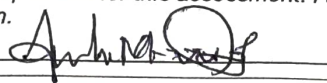
Course Code:	CSA0311	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100

CO1 Statement: Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.

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Section / Batch:	CSE.	Date:	13.07.2026

Code of Conduct

I ANLO MELWIN. S/192511299 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

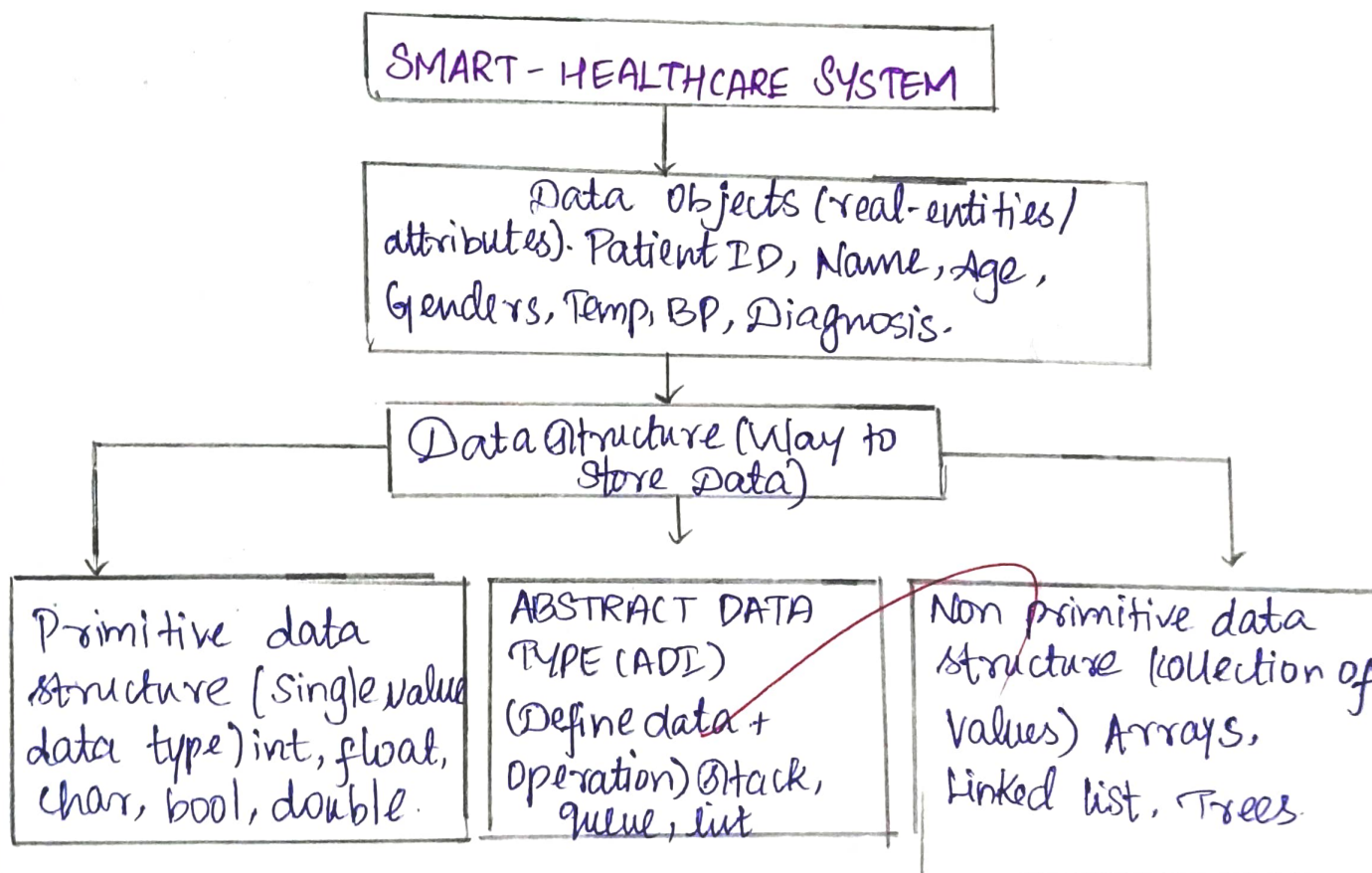
- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

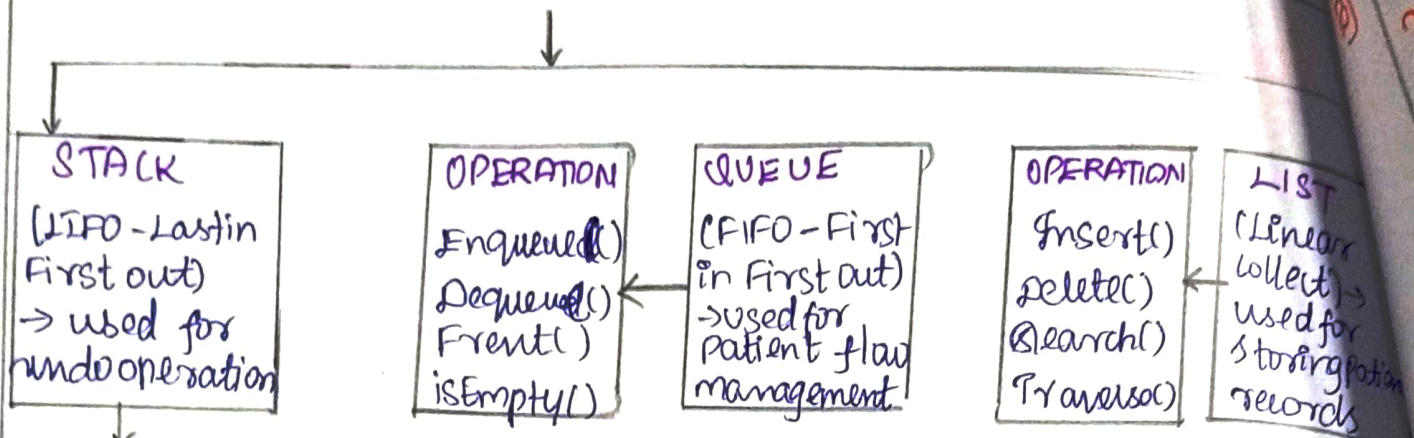
Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						


13/7/26

Q1] A Smart health care system processes large volumes of patient data such as temperature readings, medical history and diagnostic details. Construct a concept map linking the progression from Data $\rightarrow$ Data Object $\rightarrow$ Data Structure $\rightarrow$ Abstract Data Type (ADT) $\rightarrow$ Implementation in C. Identify relevant sub-concepts such as structures, pointers and operations, and establish meaningful relationships among them. Organize the concept map hierarchically and justify how level contributes to efficient data organization, processing and improved system performance.

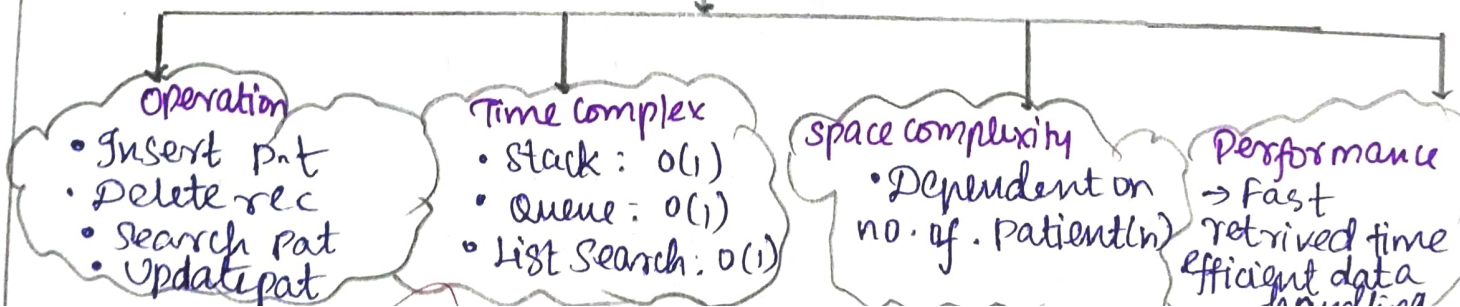




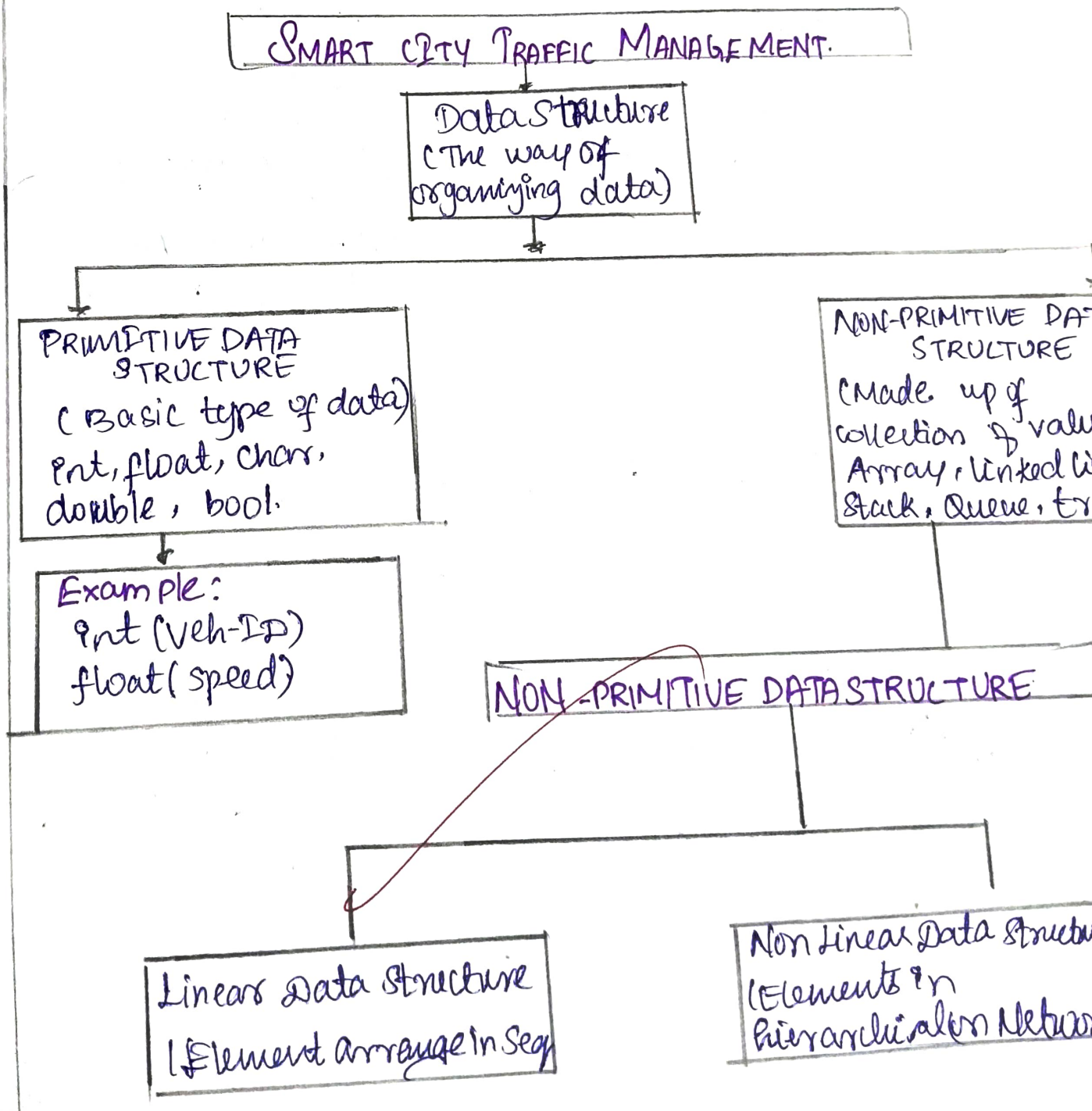
Implemented Using

Implementation In C - language			
Structure	Pointers	Functions	Dynamic Memory
struct Patient { int id; char name[50]; int age; float temp; ... };	used to link nodes in linked list / Tree struct node next;	• Insert() • Delete() • Search() • Update() • Display()	• malloc() • calloc() • free()

Leads to



3] A smart city traffic management system handles data related to vehicle movement, road networks, and signal control. Construct a concept map connecting Primitive Data Structures → Non Primitive Data Structures → Linear → Non Linear Structures. Includes examples such as array, linked lists, trees, and graphs and illustrate their applications in traffic management. Explain the relationships between these structures and justify their classification based on functionality and efficiency.

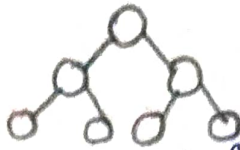


ARRAY



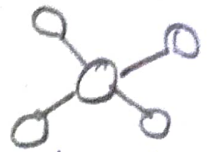
→ Fixed size
of similar element
eg: `count[10];`

TREE



→ Hierarchical
Structure.
eg: Road hierarchy

GRAPH



→ Network of
node connect by
edges.

Linked List

→ Dynamic collection
of elements
eg: Store list of
vehicles detect by
Sensors.

Queue.



(FIFO)

Used for waiting
lines.

Eg: vehicles wait
at traffic signals

APPLICATIONS IN SMART CITY TRAFFIC.

VEHICLE MONITORING

• Store and
process vehicle
data from
sensor

TRAFFIC SIGNAL CONTROL

• Queue data
helps signal
scheduling

ROUTE OPTIMIZATION

Graph help in
find shortest
/ fastest.

CONGESTION DETECTION

Analyze
traffic
density
using array
& graphs.